

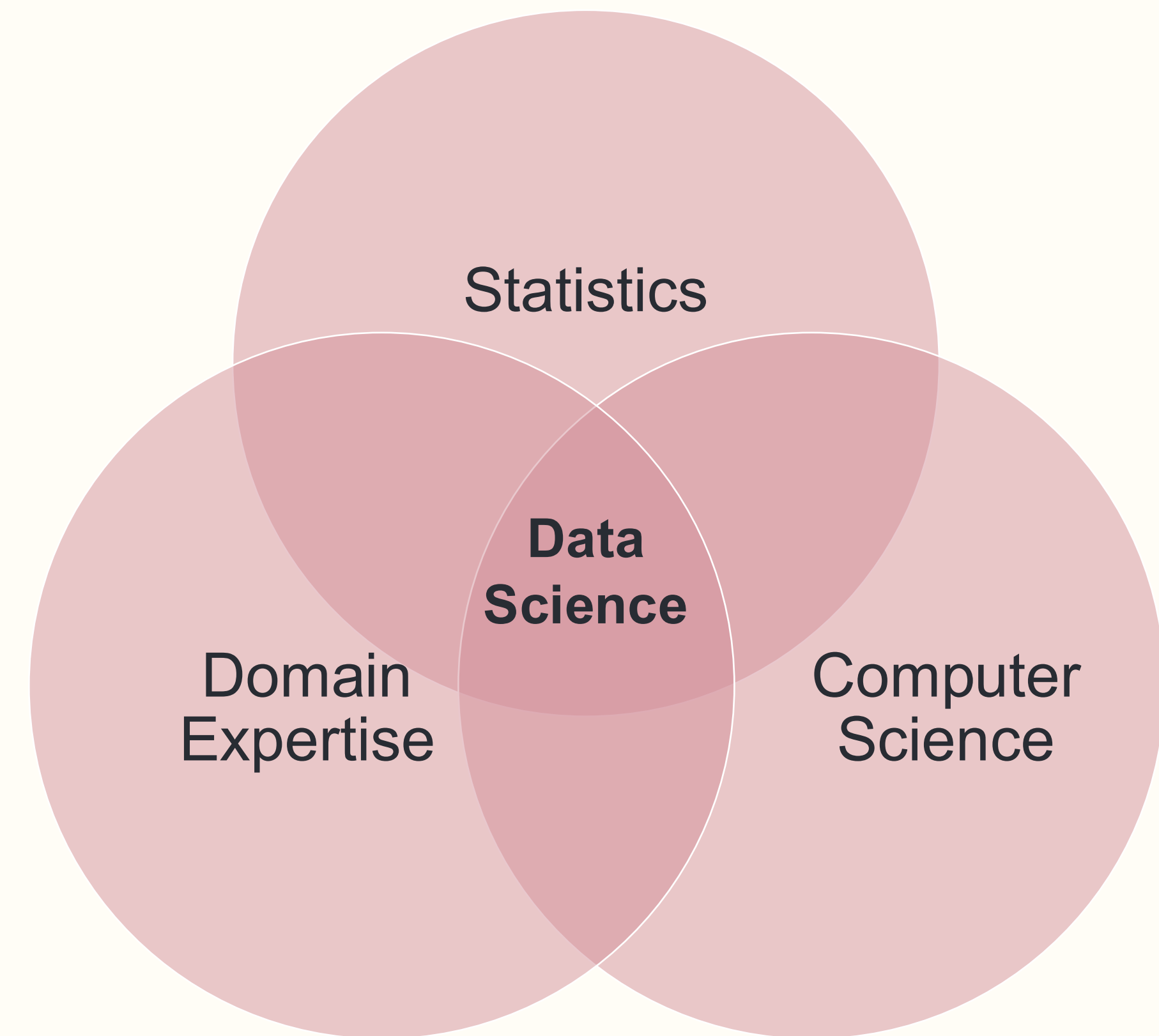
Lecture 1: Introduction to course, R, RStudio, and R Markdown

Heather Mattie, PhD

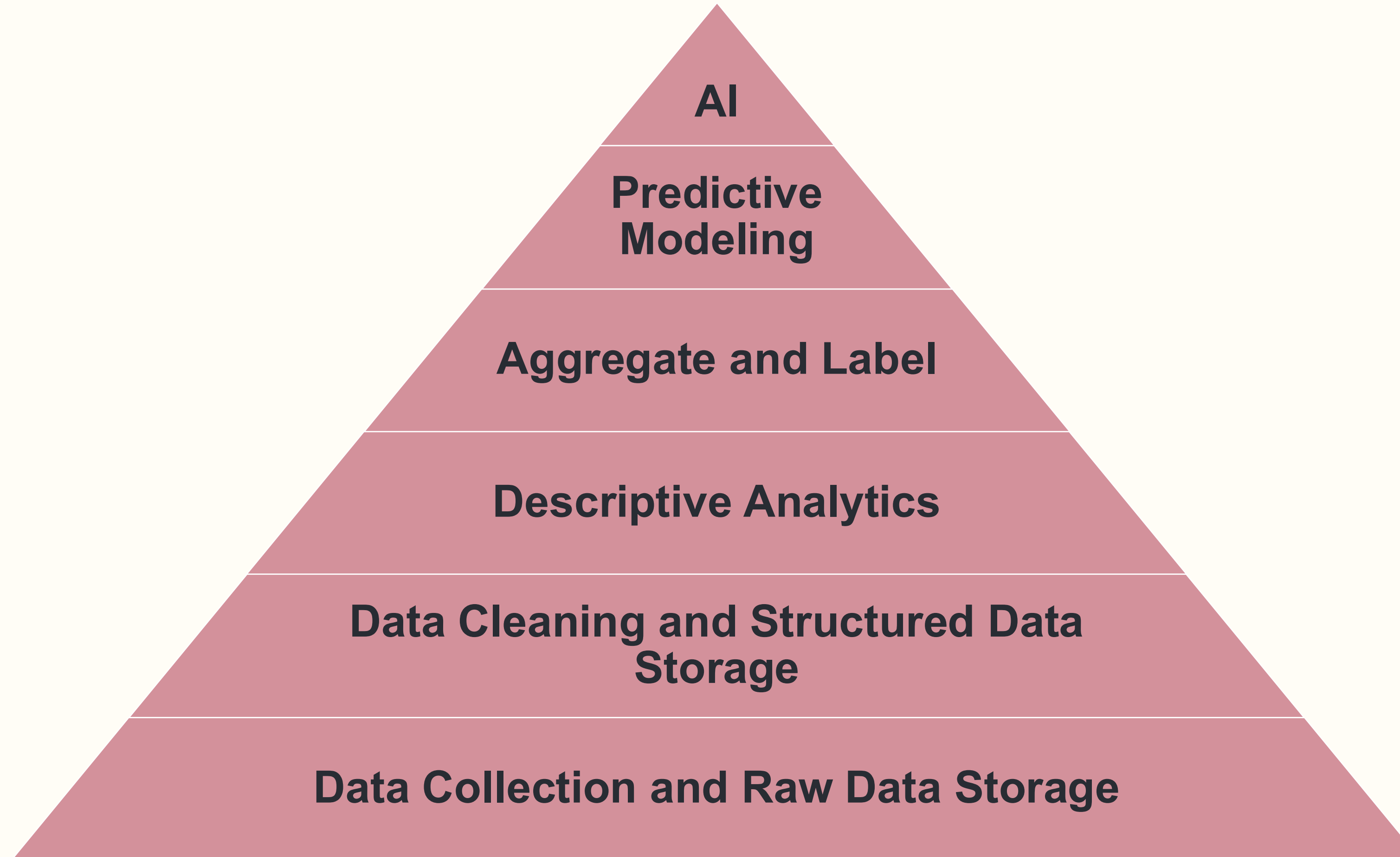
September 1, 2026

What is Data Science?

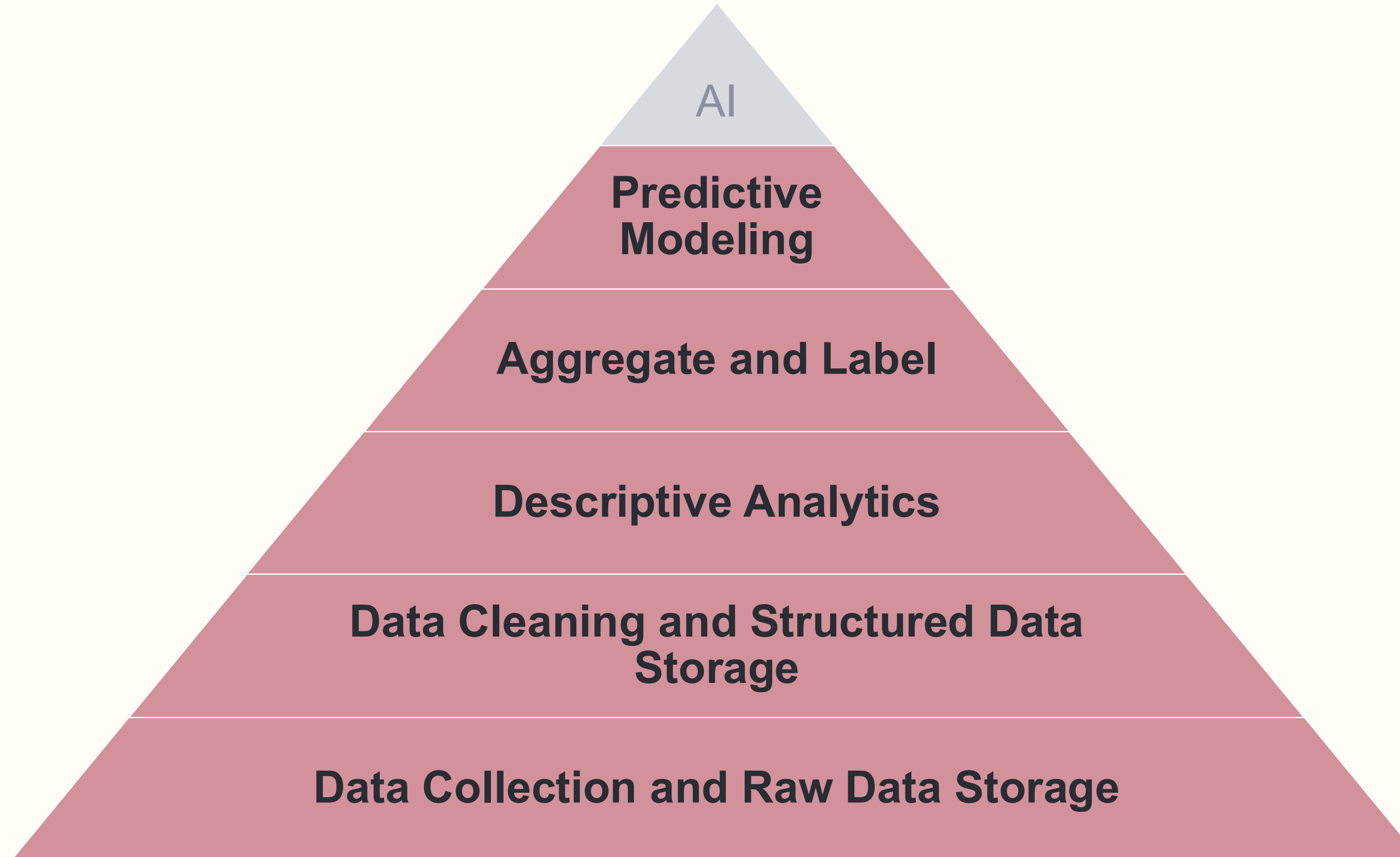
Data Science is a **multidisciplinary** field that uses scientific methods to **extract knowledge** and **insights** from **data**



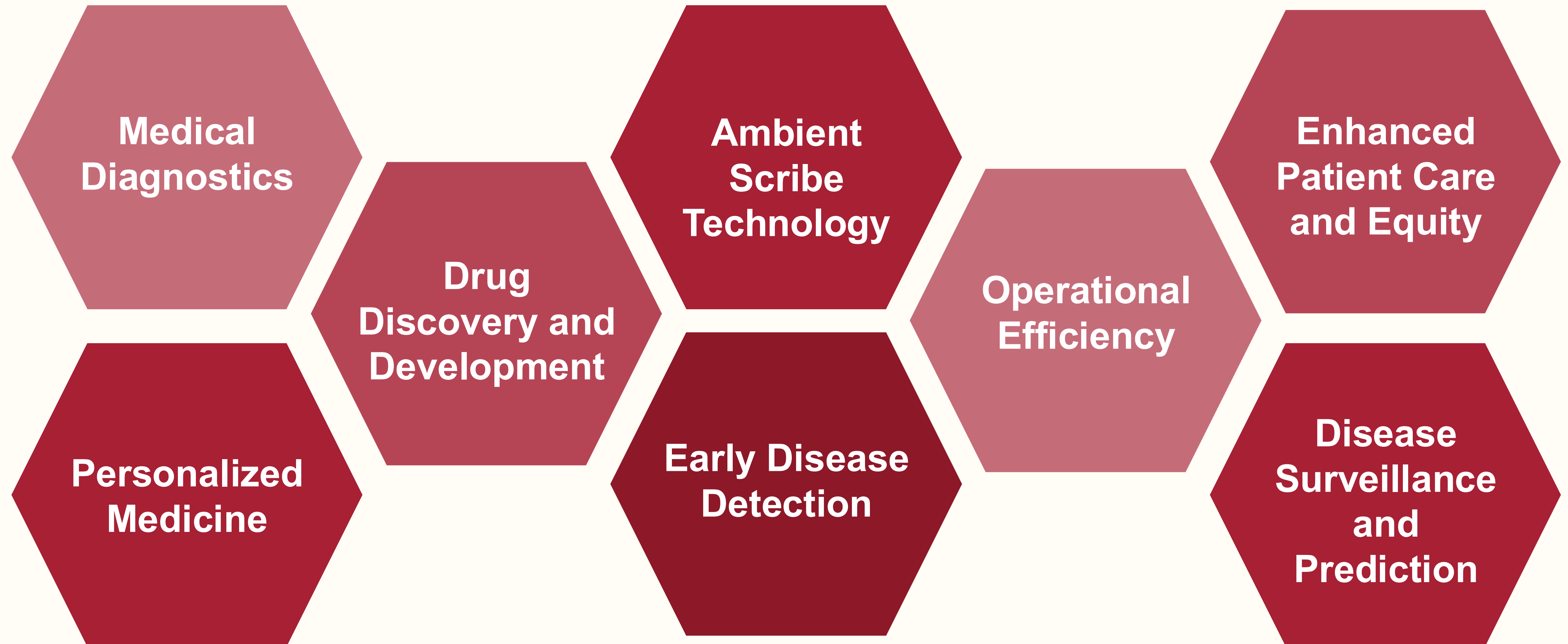
Data Science Hierarchy of Needs



Data Science Hierarchy of Needs



Data Science in Public Health – Successes



Data Science in Public Health – Pitfalls

Biased Algorithms

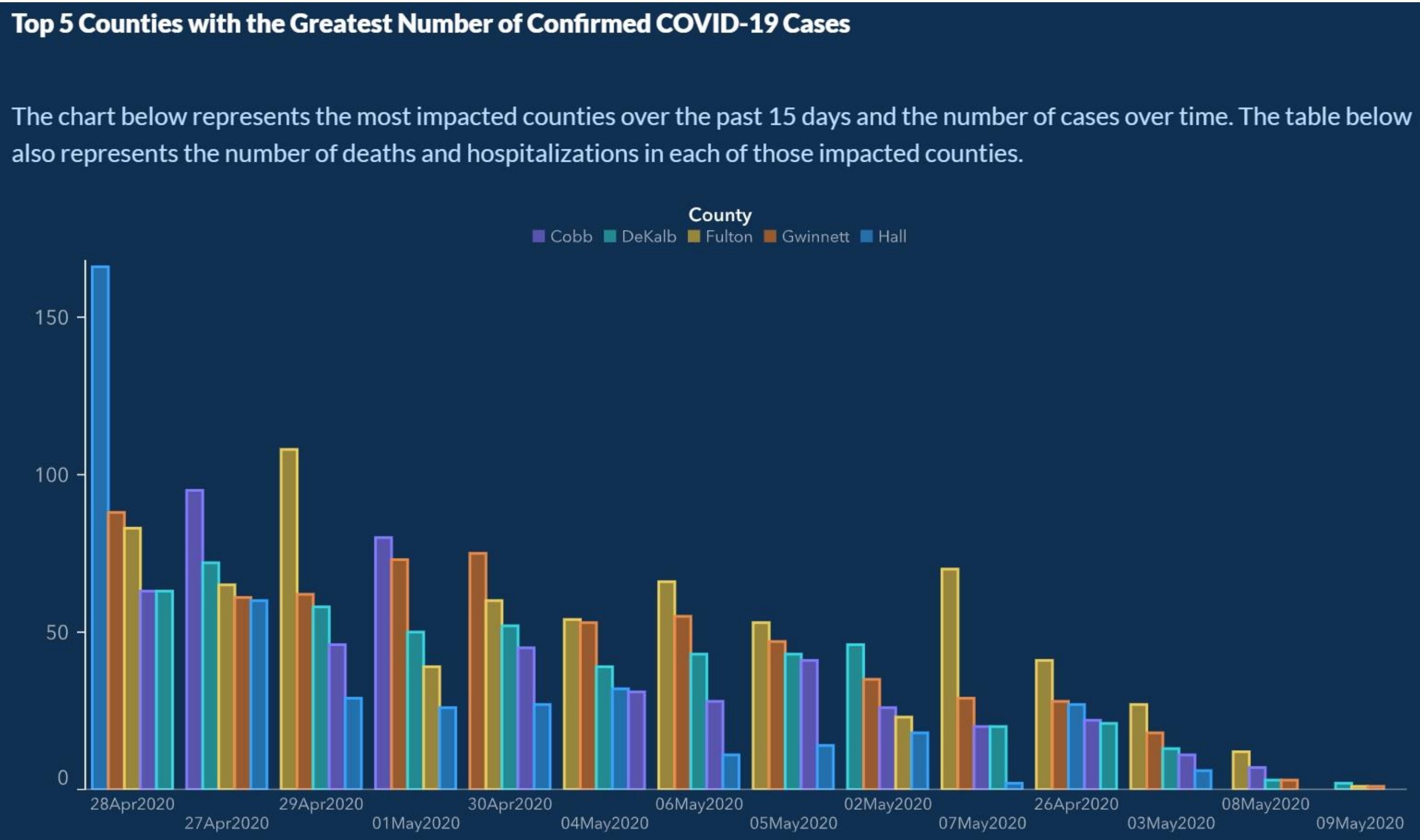
Data Privacy and Security Risks

Lack of Transparency

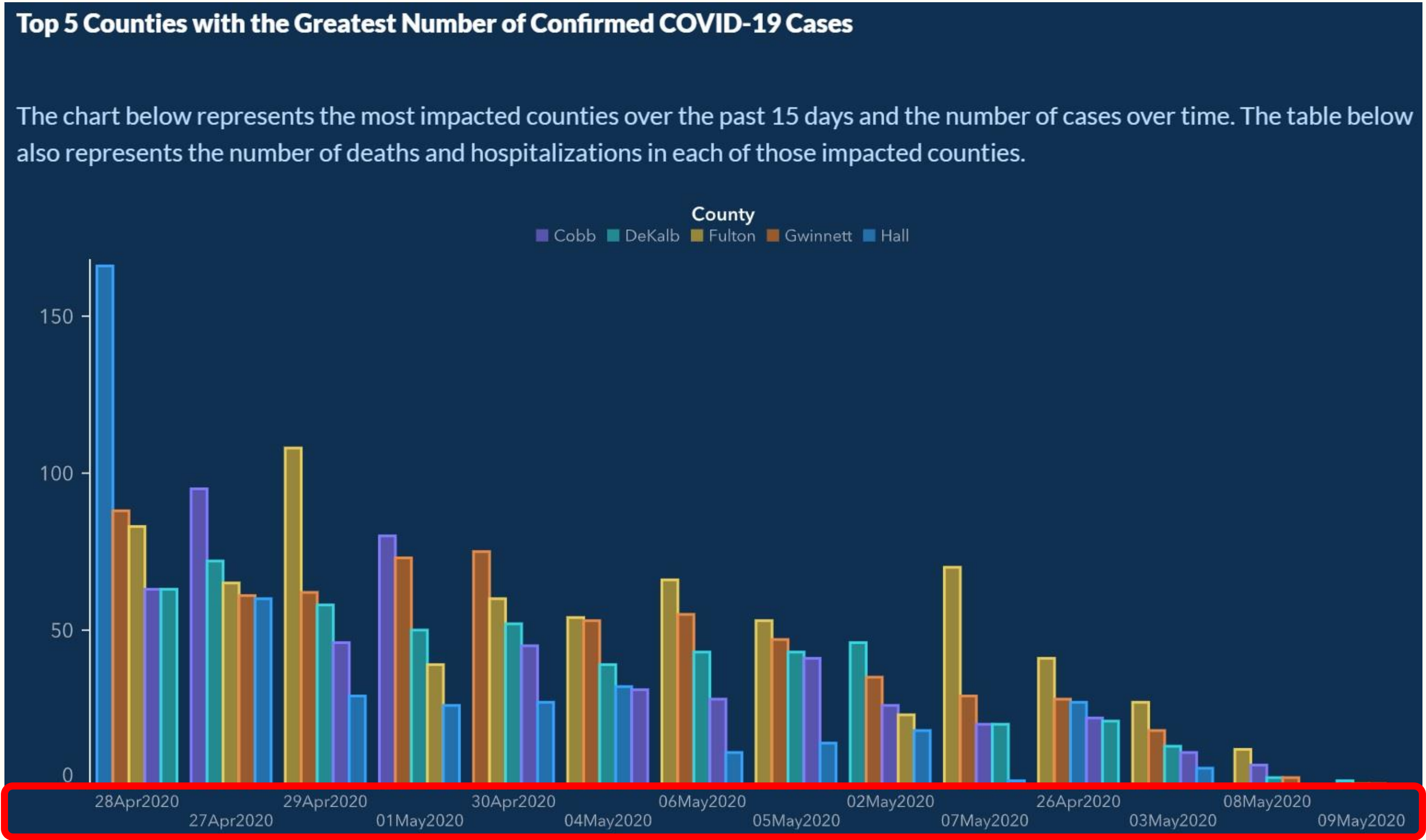
AI Regulation Landscape

Mis- and Disinformation

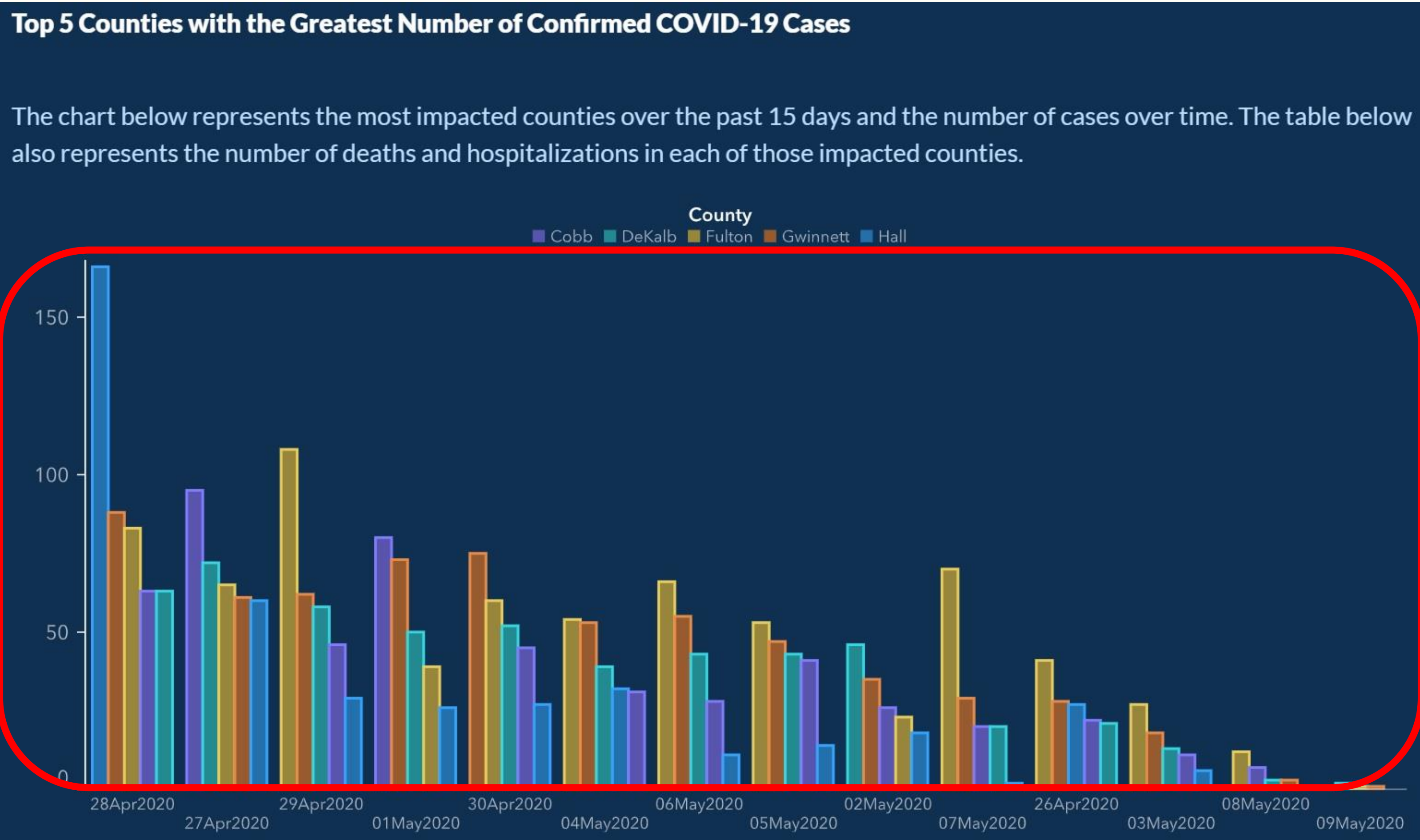
Misinformation Example



Misinformation Example



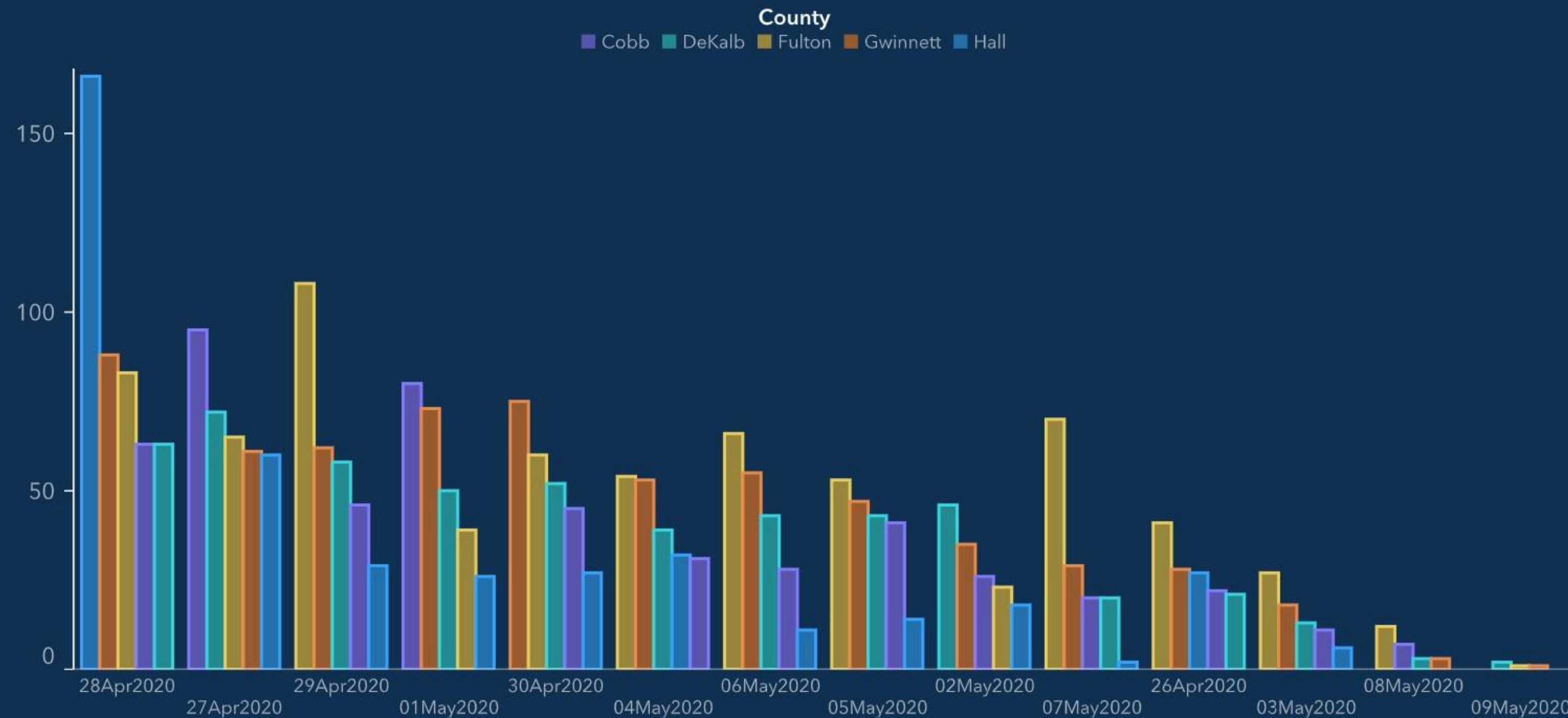
Misinformation Example



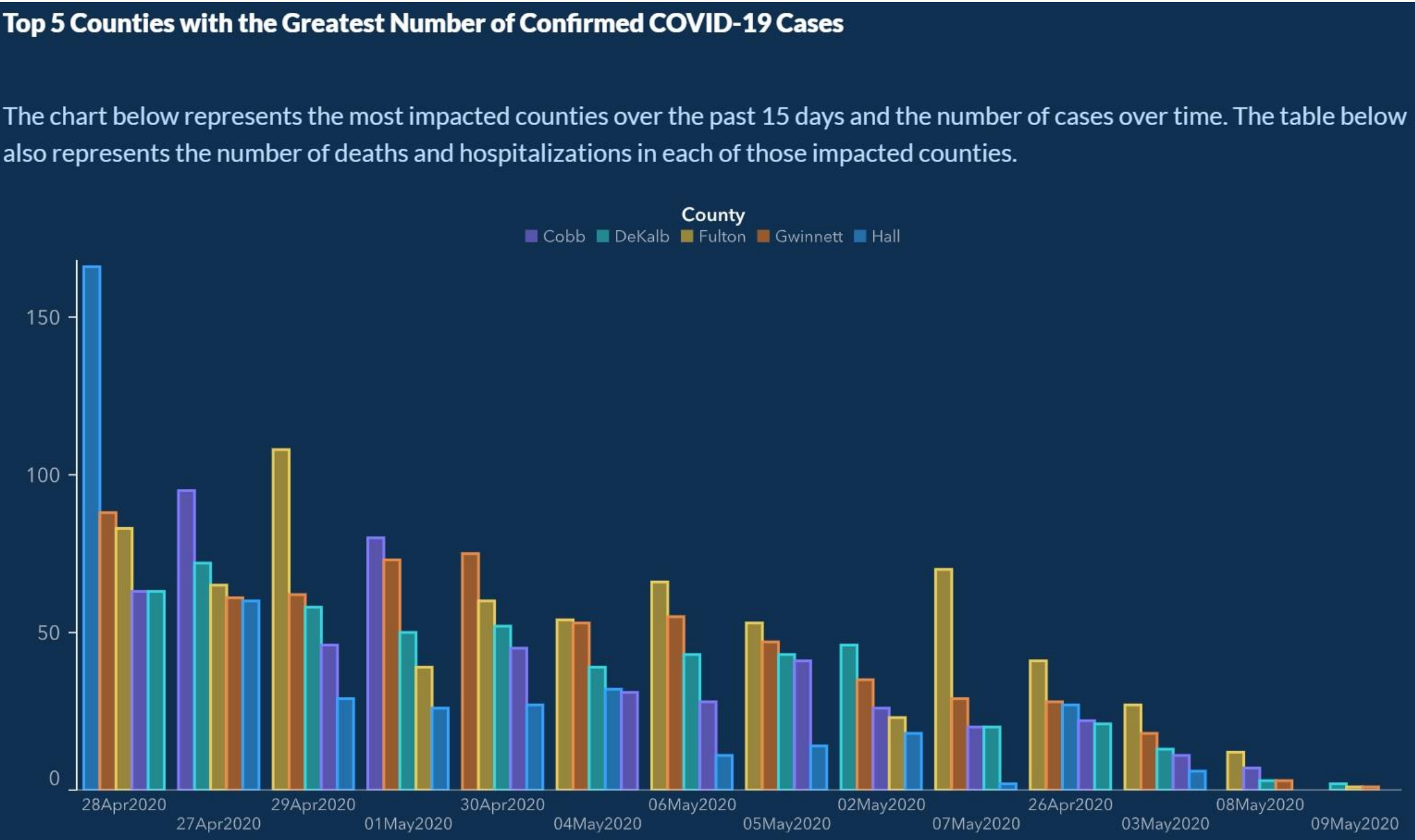
Misinformation Example

Top 5 Counties with the Greatest Number of Confirmed COVID-19 Cases

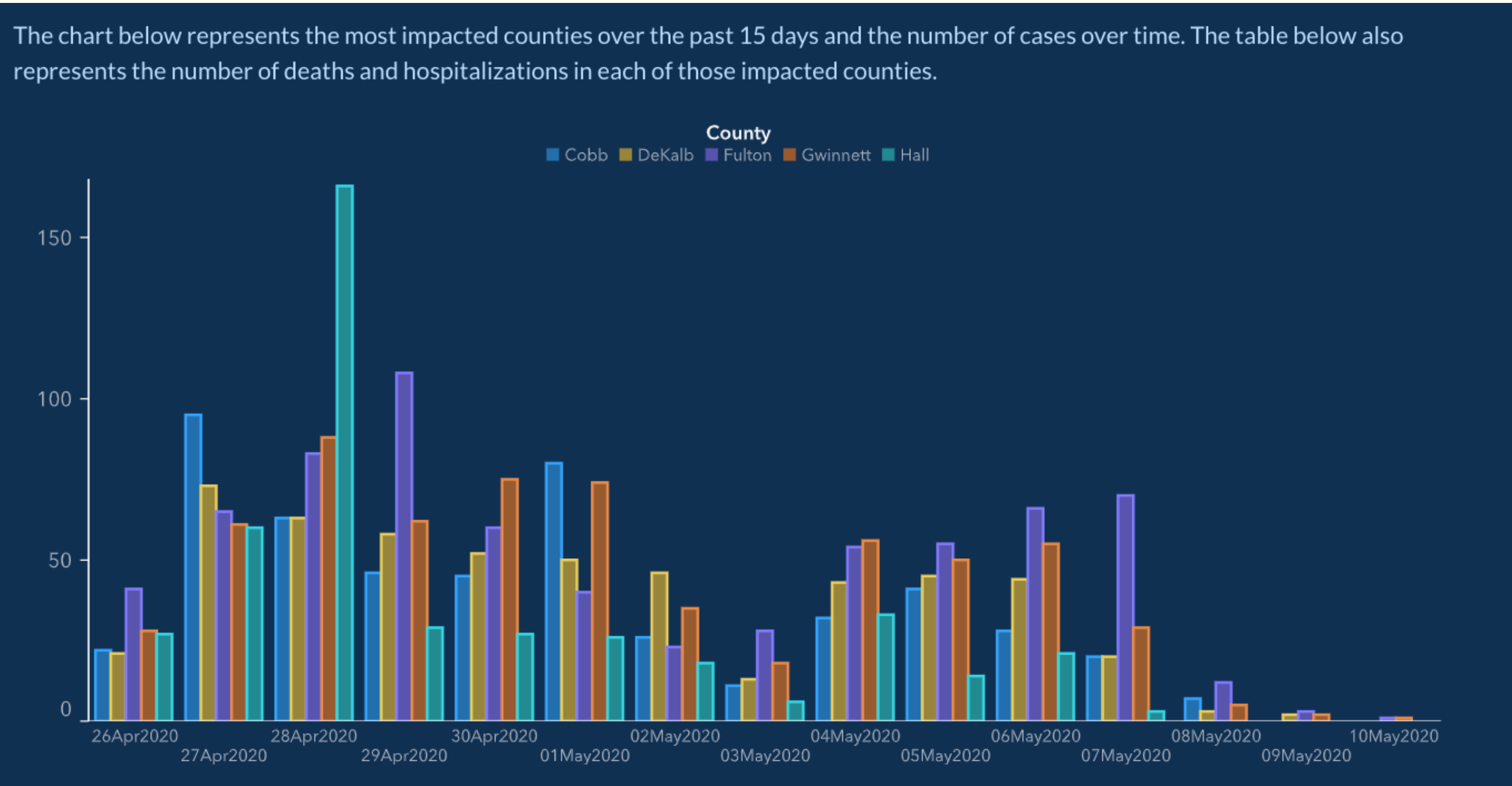
The chart below represents the most impacted counties over the past 15 days and the number of cases over time. The table below also represents the number of deaths and hospitalizations in each of those impacted counties.



Misinformation Example



Corrected version



As data
[scientists, analysts, creators, consumers],
we all play a role in **understanding** and **confronting**
ethics, equity, and bias in data, algorithms, clinical
AI, and healthcare

The Data Science Pipeline



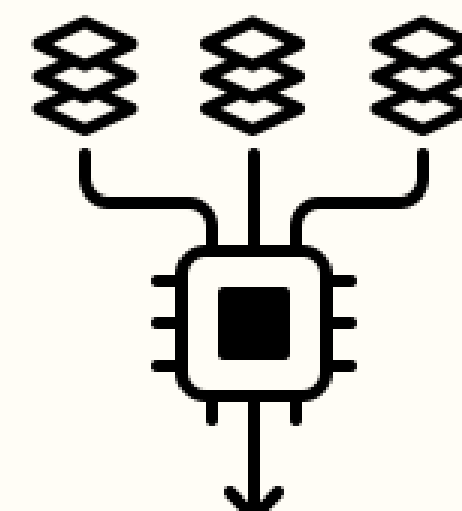
**Collecting
and
importing
(loading)
the data**



**Processing
(cleaning,
wrangling)
the data**



**Visualizing
and
summarizing
the data**



**Building
models
(statistical
and ML)**



**Interpretation
and
communication
of results**

Who is helping you learn data science?

Instructor

Heather Mattie
Lecturer on Biostatistics
hemattie@hsph.harvard.edu



Teaching Fellows

Stephanie Armbruster
sarmbruster@g.harvard.edu

Monica Robles Fontan
mroblesfontan@g.harvard.edu

Kimberly Greco
kimberly.greco@g.harvard.edu

Office Hours and Lab

Office Hours

Day	Time	Staff	Location
Thursday	12-1 pm	Heather	Building 1, 4 th floor, room 421A
Wednesday	10:30-11:30 am	Stephanie	Building 2, 4 th floor, room 426
Friday	11:30-12:30 pm	Monica	Building 2, 4 th floor, room 426
Monday	1-2 pm	Kimberly	Zoom

Lab

Day	Time	Location
Friday	9:45-11:15 am	Zoom

***Lab will not be held every week**



Office Hours and Lab

- Heather's office hour this week will be Thursday, 2-2:30pm
- The TFs will not hold office hours this week
- There is no lab this week

Grading

Homework

- 5 assignments
- 60% of final grade
- You are welcome to discuss the course material and homework questions with others, but the work you turn in must be your own. Be sure to cite any sources you use, including generative AI.

Take-home Midterm

- 15% of final grade
- Questions that require writing code and short answers
- You are not allowed to work on or discuss this assignment with other students or use generative AI

Grading

Final Project

- 25% of final grade
- Will work in teams of 4-5 people
- Will choose 1 of 3 possible projects

Pass/Fail Threshold

- If taking course pass/fail, must earn final grade of 70% or more to pass

Auditing

- Auditors do not need to submit any assignments

Generative AI Policy

As a tool, generative AI can provide valuable assistance in understanding concepts, troubleshooting issues, or even offering general guidance (use discretion!). However, to ensure true learning in an introduction to data science course, you should not rely on generative AI to complete your homework/midterm/final project. Make sure you understand the course content. **See the syllabus for more details on the generative AI policy.**

Homework Assignments

- Real-world/public health/medical focus
- Scrape and wrangle/clean messy data
- Explore data
- Visualize data
- Perform statistical analyses
- Make predictions (build ML models)
- Communicate results
- Will be written in R using R Markdown and submitted via Canvas
- Must also submit knitted html file
- Points will be deducted if we are unable to knit your R Markdown file when grading
- Can use **2 late days per assignment**

Midterm Exam

- You will have 1 week to complete the midterm exam
- Very similar to a homework assignment
 - Similar length
 - Submit via Canvas
 - R Markdown and html files must be submitted
- No collaboration of any kind is permitted
- Use of generative AI is not permitted

Final Project

- Teams of 4-5 students
- Choose a project from list of 3
 - Will be given a dataset and prompt and will be tasked with data wrangling, data visualization, statistical analyses, building ML models and communicating your findings
- A TF will be assigned to each team to give advice throughout the project
 - Assigned in beginning of November

Course Materials

- 1. Lecture slides:** condensed content and main concepts
 - 2. R Markdown (.Rmd) files:** complete text explanations and example code shown in class, plus self-assessment questions and solutions and FAQs at the bottom
- All course material is available on Canvas organized by week and [GitHub](#) organized by topic (module)

Course Communication

- Email me or the teaching fellows
- Do **not** send a Canvas message – I rarely check them!
- Office hours
- Lab sessions
- See me during the class break or after lecture

Course Expectations

- You are **highly** encouraged, but not required, to attend lecture
 - Each lecture will be recorded and available on the Canvas course site
- Participation is not required or included as part of your final grade, but is highly encouraged
 - We all learn from each other
 - After a while I start to dislike the sound of my own voice
- Attending a weekly lab session is highly recommended but not required
 - The session will be recorded and available on the Canvas course site
- Break time - we'll take a 5- minute break around the middle of each lecture (45-50 minutes in)

Course Conduct: Chatham House Rule

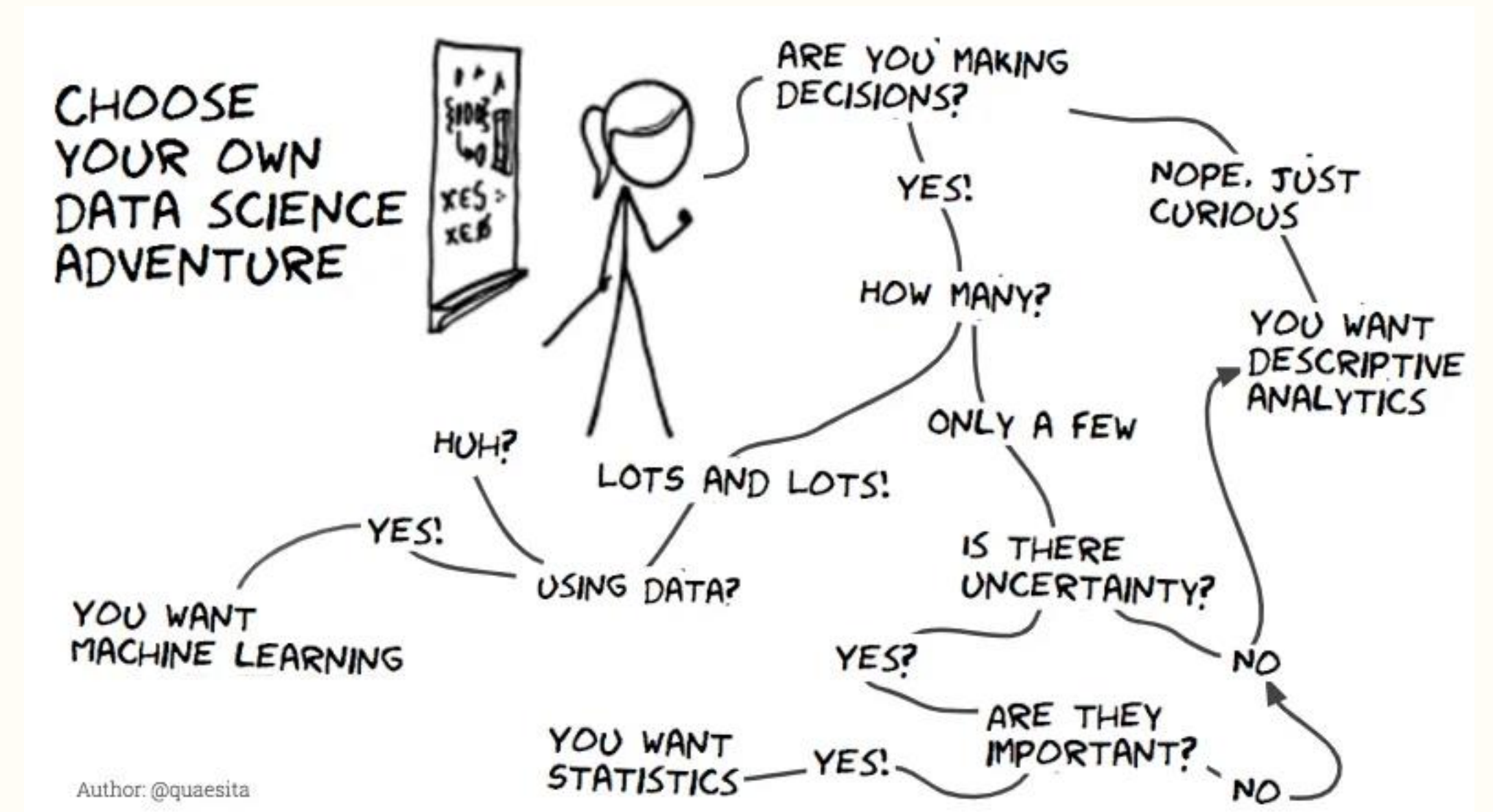
To encourage openness, trust and candid discussion across a variety of topics at Chan, I propose we adopt a **Non-Attribution Community Norm** for this course, modeled after the Chatham House Rule. This social contract means you may share the ideas discussed in class outside of it, but you may not identify who said them or what affiliation they hold.

This means:

- You are welcome to talk about class ideas outside of class.
- Do not quote classmates by name outside of class or on social media.
- Screenshots, recordings, or transcripts may **not** be circulated without written permission from faculty, speakers, etc.
- The rule also applies in breakout rooms (either online or in person).

This course IS...

- An introduction to R and RStudio
- An introduction to creating publishable visualizations using `ggplot2`
- An introduction to using git and GitHub for version control and reproducibility
- Focused on foundational knowledge for coding, the data science project pipeline, machine learning, and communication of process and results
- A chance for you to collaborate with and learn from other students
- A safe space to “get your hands dirty” and write and debug code



This course is NOT...

- A machine learning course
 - You will get a great foundation in ML from this course, but ML is only about 40% of the course
- Meant to be a silver bullet for all data science tasks / projects
 - Passing this course does not mean you won't have more to learn - data science is constantly evolving and we need to be lifelong learners

Action Items

- Download and install [R](#) and [RStudio](#)
 - Make sure you **download R first**
- Create a [GitHub account](#)
 - If you are interested in downloading course materials from GitHub
- Complete this [survey](#)

Introduction to R, RStudio, and R Markdown

Introduction to R

Heather Mattie, PhD

What is R?

R is a programming language and software environment specifically designed for **statistical computing** and **graphics**

- Open source
- Specialized for statistics
- Active community
- Reproducible research



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Download

[CRAN](#)

R Project

[About R](#)

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[Contributors](#)

[What's New?](#)

[Reporting Bugs](#)

[Conferences](#)

[Search](#)

[Get Involved: Mailing Lists](#)

[Get Involved: Contributing](#)

[Developer Pages](#)

[R Blog](#)

The R Project for Statistical Computing

Getting Started

R is a free software environment for statistical computing and graphics. It compiles and runs on a wide variety of UNIX platforms, Windows and MacOS. To [download R](#), please choose your preferred [CRAN mirror](#).

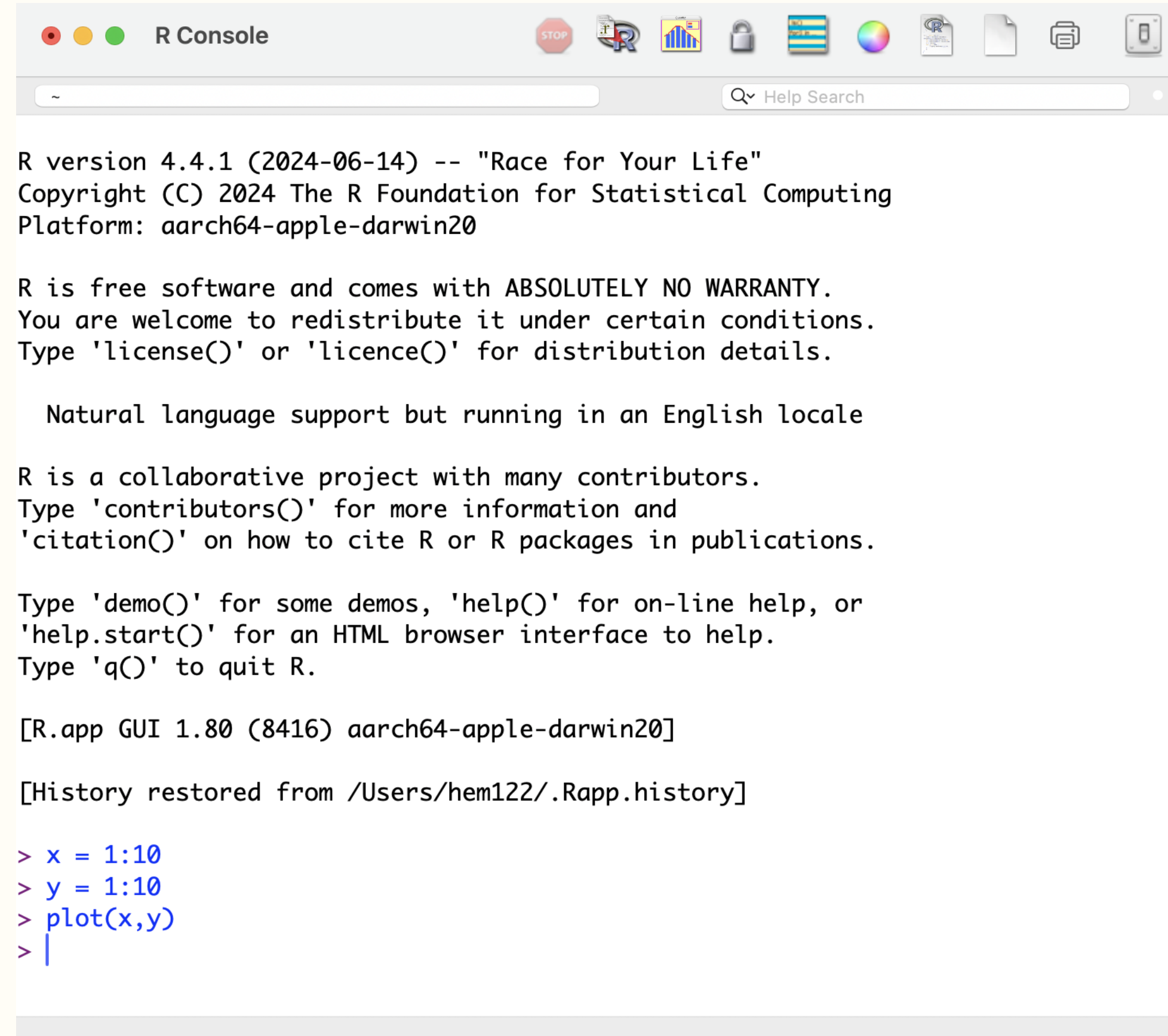
If you have questions about R like how to download and install the software, or what the license terms are, please read our [answers to frequently asked questions](#) before you send an email.

News

- **R version 4.5.1 (Great Square Root)** has been released on 2025-06-13.
- **R version 4.5.0 (How About a Twenty-Six)** has been released on 2025-04-11.
- **R version 4.4.3 (Trophy Case)** (wrap-up of 4.4.x) was released on 2025-02-28.
- The **useR! 2025** conference will take place at Duke University, in Durham, NC, USA, August 8-10.
- We are deeply sorry to announce that our friend and colleague Friedrich (Fritz) Leisch has died. [Read our tribute to Fritz here.](#)



What is R?



```
R version 4.4.1 (2024-06-14) -- "Race for Your Life"
Copyright (C) 2024 The R Foundation for Statistical Computing
Platform: aarch64-apple-darwin20

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

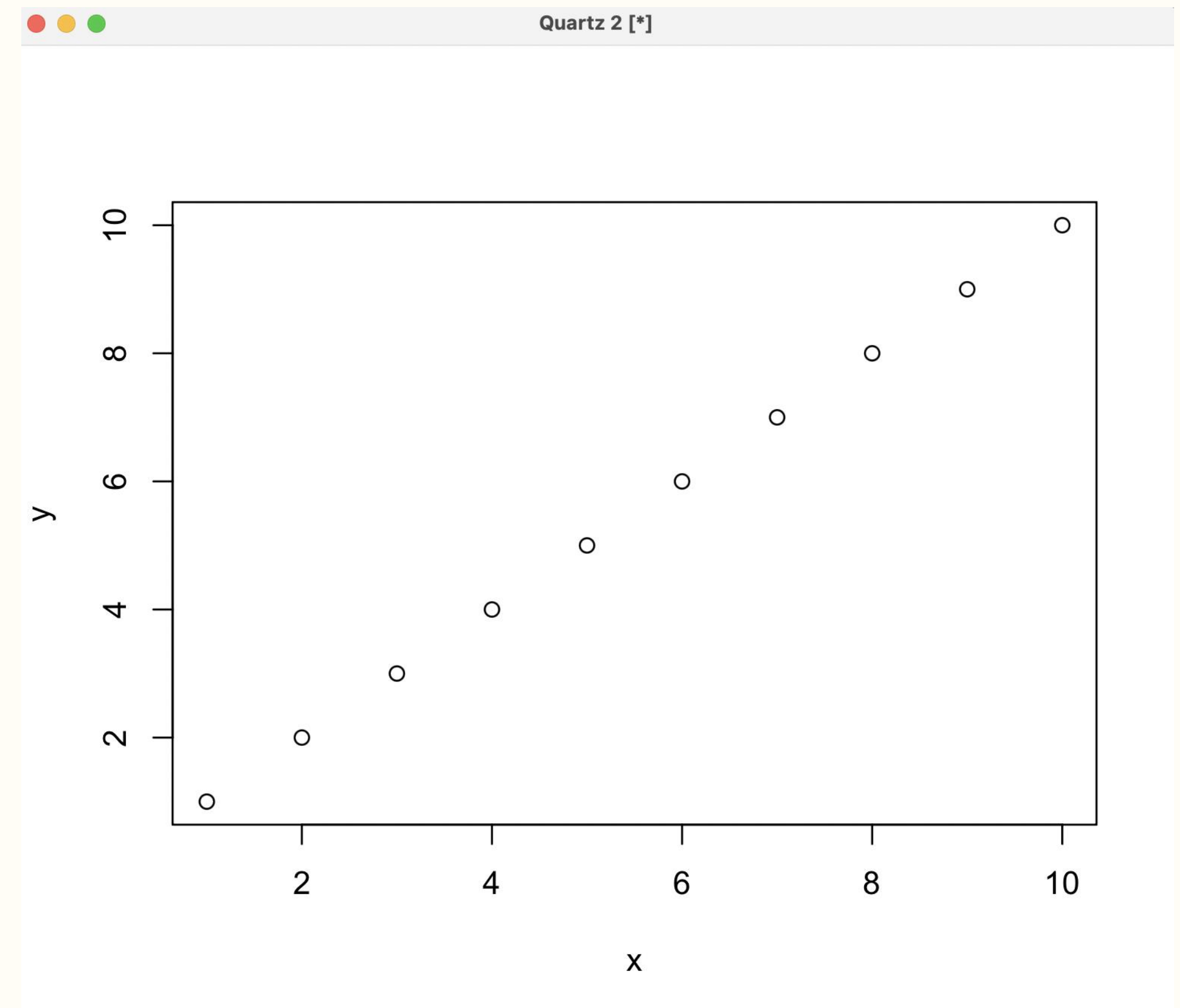
Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[R.app GUI 1.80 (8416) aarch64-apple-darwin20]
[History restored from /Users/hem122/.Rapp.history]

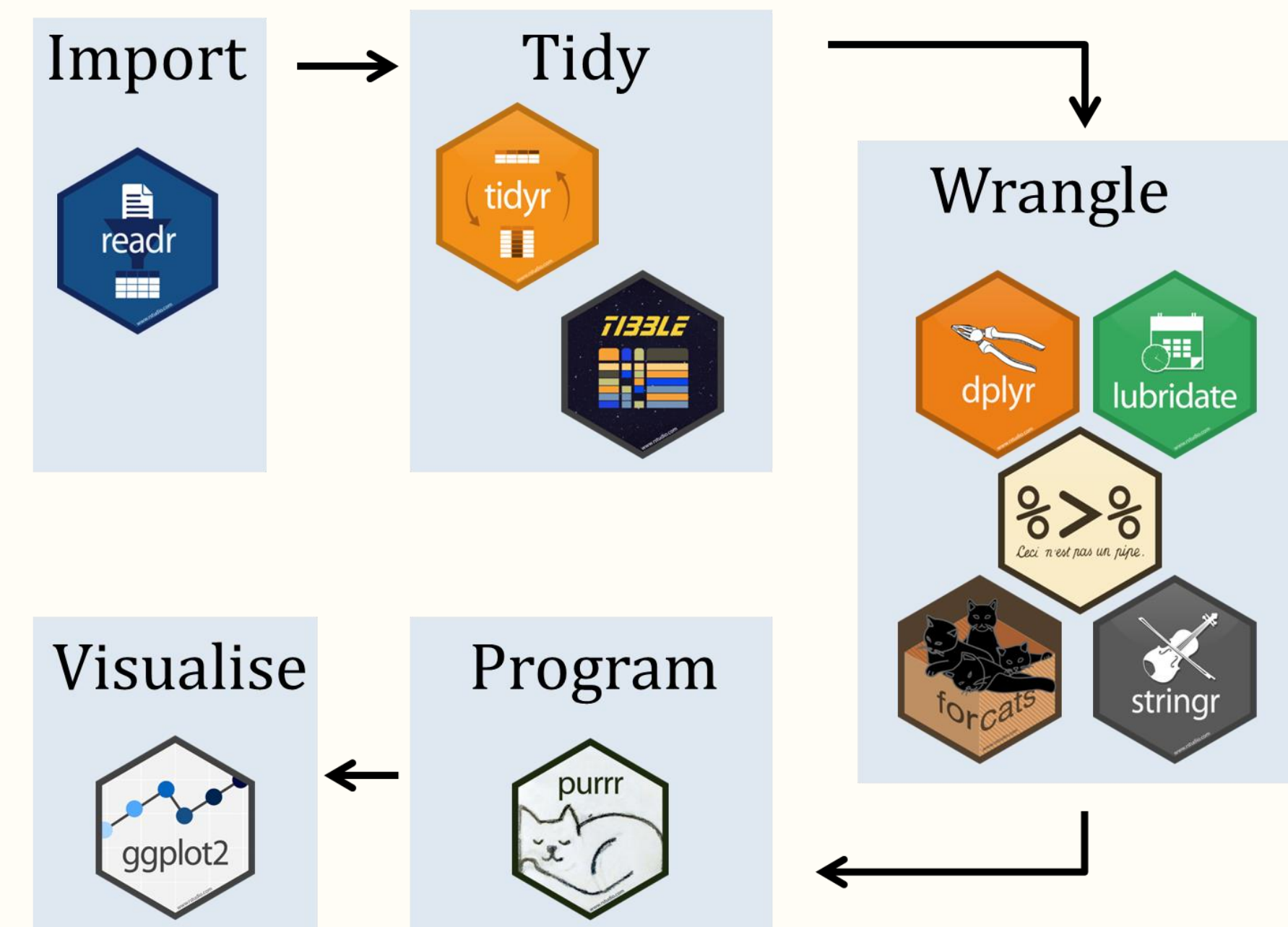
> x = 1:10
> y = 1:10
> plot(x,y)
> |
```



Packages

In R, **packages** are collections of functions, data, and documentation that extend the base functionality of the language.

- Allow users to perform specific tasks or analyze particular types of data
- Essentially bundles of reusable code that can be easily shared and installed
- Must first install a package, and then load it when you want to use it
- All have documentation available online



The R Community

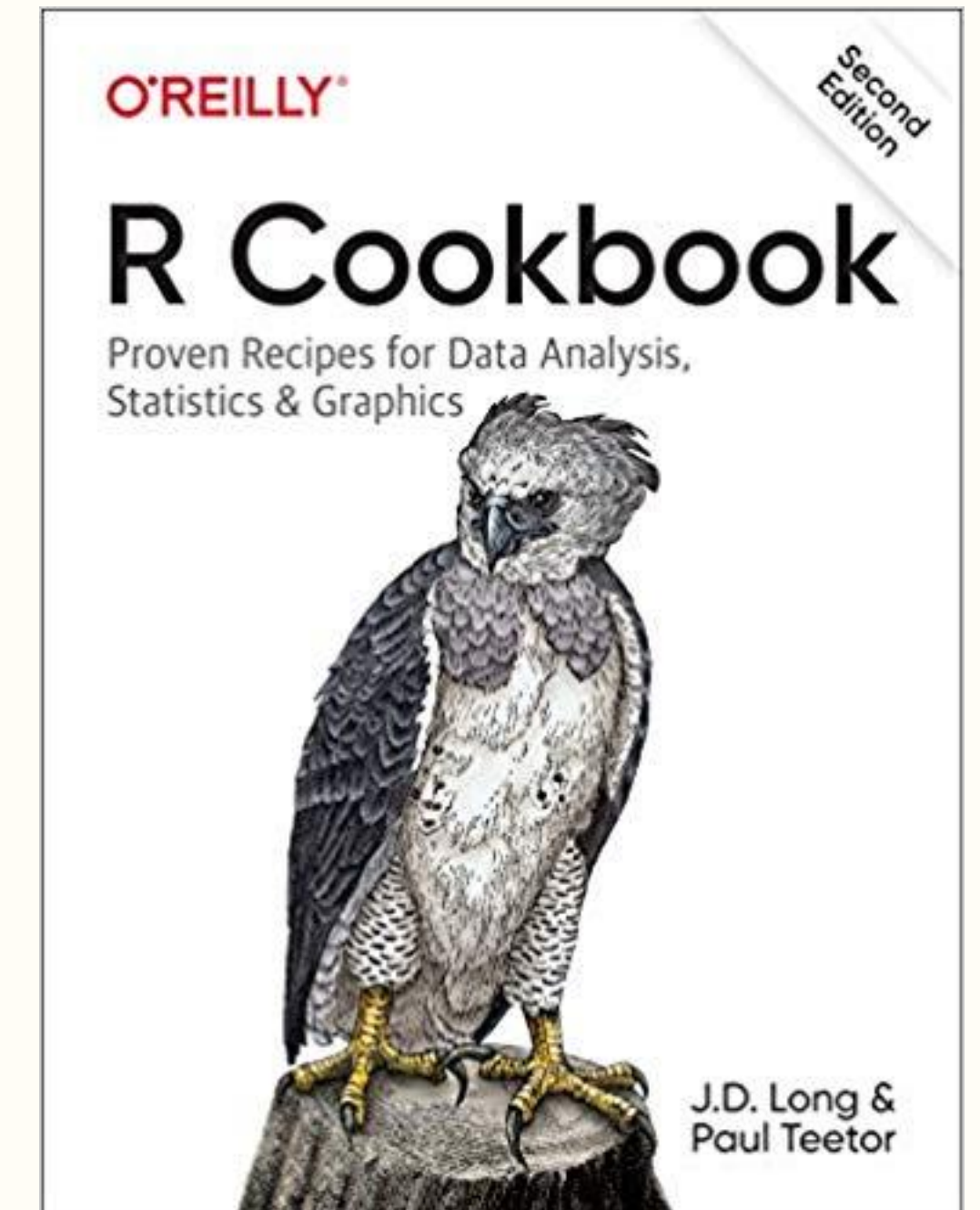
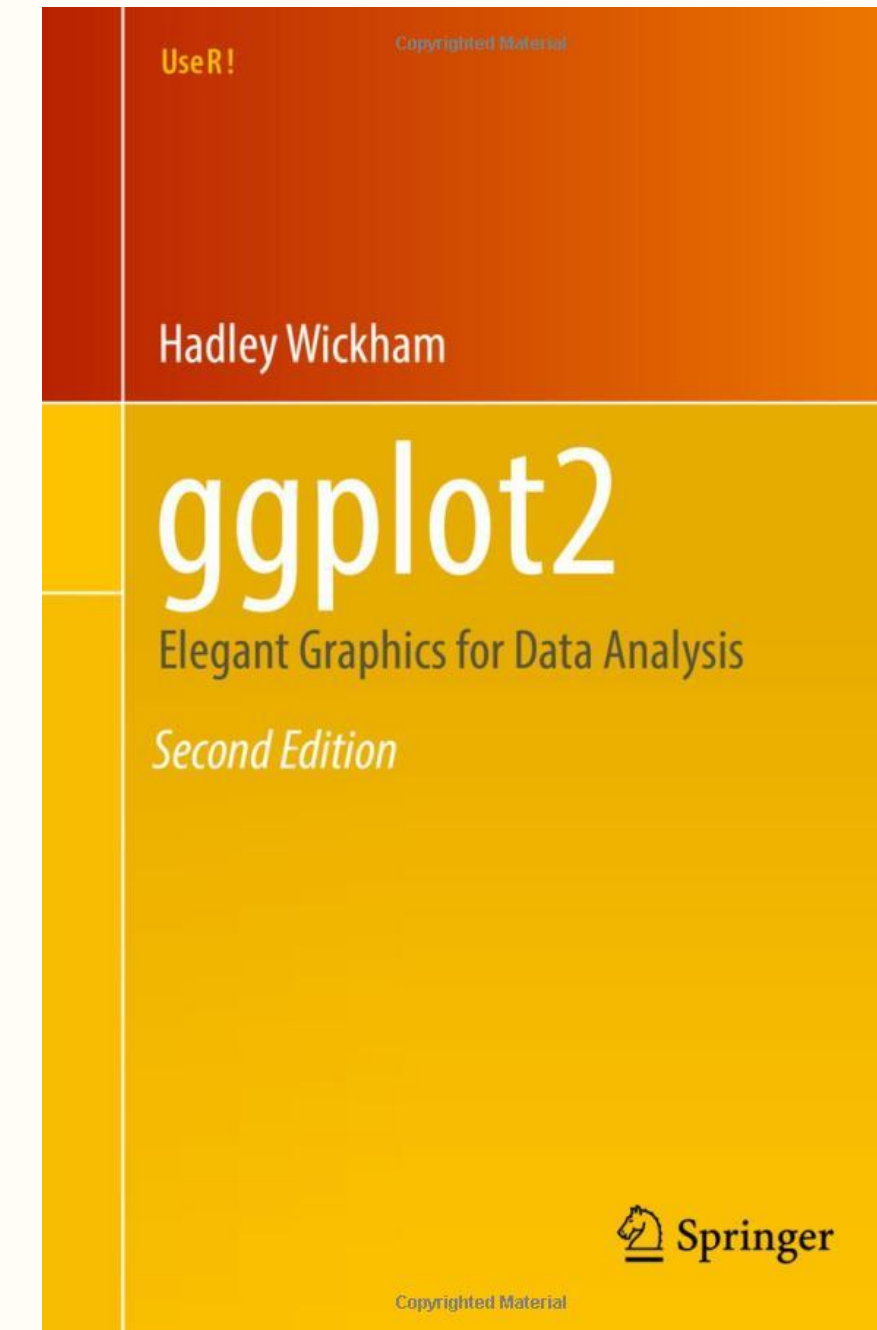
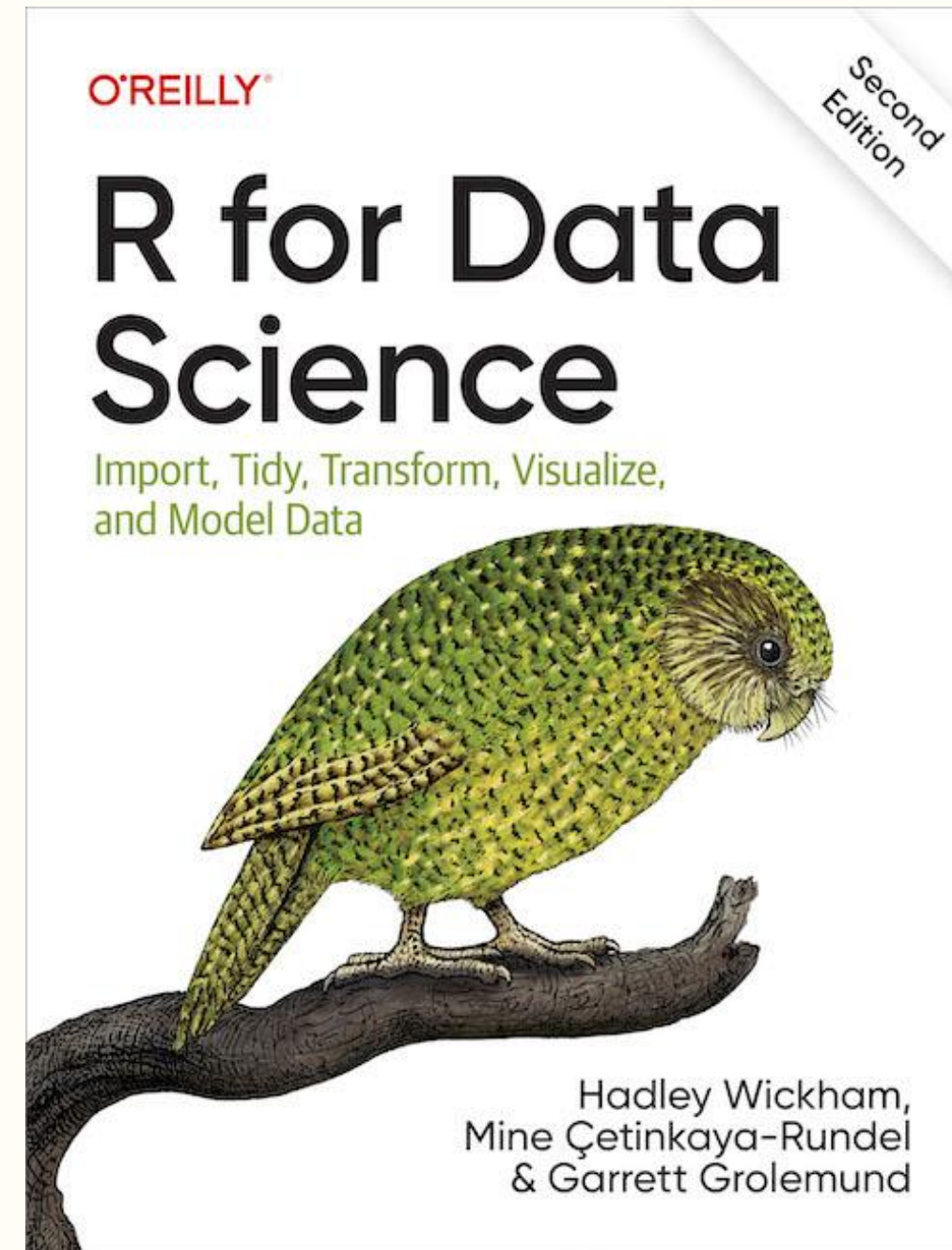
One of the best things about R is the **R community**. The R Community is a global network of individuals who use, contribute to, and promote the R programming language.

- Active engagement through online forums, blogs, social media, and conferences
- Members range from beginners to experts in various fields
- The community fosters collaboration, knowledge sharing, and support for users of all levels

R Resources

For help writing code:

- R for Data Science
- ggplot2
- R Cookbook



R Resources

For help **debugging** code:

- R Help: `help()` and ?
- Vignettes and code demonstrations
- Package documentation
- Stack Overflow

stackoverflow About Products OverflowAI [r]

Home Questions Tags Challenges NEW Chat Articles Users Jobs Companies

COLLECTIVES + Communities for your favorite technologies. [Explore all Collectives](#)

[r]

R is a free, open-source programming language and software environment for statistical computing, bioinformatics, information graphics, and general computing.

Sign up to watch this tag and see more personalized content

Watch tag Go to Wiki

510,956 questions Newest Active Bountied Unanswered More Filter

1 vote Method to set minimum column width in flextable

0 answers I'm looking for a way to apply `autofit()` only to columns that are too small. This is my current method but I'm looking for a better way to do it: `library(flextable) library(magrittr) ### Ex. Tbls ...`

22 views R r r-flextable Aaron Rose 11 asked 3 hours ago

-1 votes Extract last numbers from text strings [duplicate]

ie with a column of strings that have either 1 or 2 numbers at the end of the
:t these 1 or 2 numbers and place them in a new column. Here is how I am ...

jr Eric Krantz 2,257 asked 3 hours ago

age related to geocoding that could help me assign cate-
closed]

project where I am trying to compile polygons of various Parks at different
national, etc.) In my dataset, I only have data on the name of each Park, an...

ing reverse-geocoding sam 1 asked 6 hours ago

Package ‘dslabs’

March 1, 2024

Title Data Science Labs

Version 0.8.0

Description Datasets and functions that can be used for data analysis practice, home-work and projects in data science courses and workshops. 26 datasets are available for case studies in data visualization, statistical inference, modeling, linear regression, data wrangling and machine learning.

Author Rafael A. Irizarry, Amy Gill

Maintainer Rafael A. Irizarry <rafael_irizarry@dfci.harvard.edu>



Summary

- R is a programming language designed for statistical computing and visualizations
- R is open source and has a large, active, and diverse community that contributes to and promotes R
- R packages are collections of functions, data, and documentation that extend the base functionality of R
- There are many resources for writing and debugging R code

Introduction to R, RStudio, and R Markdown

Introduction to RStudio

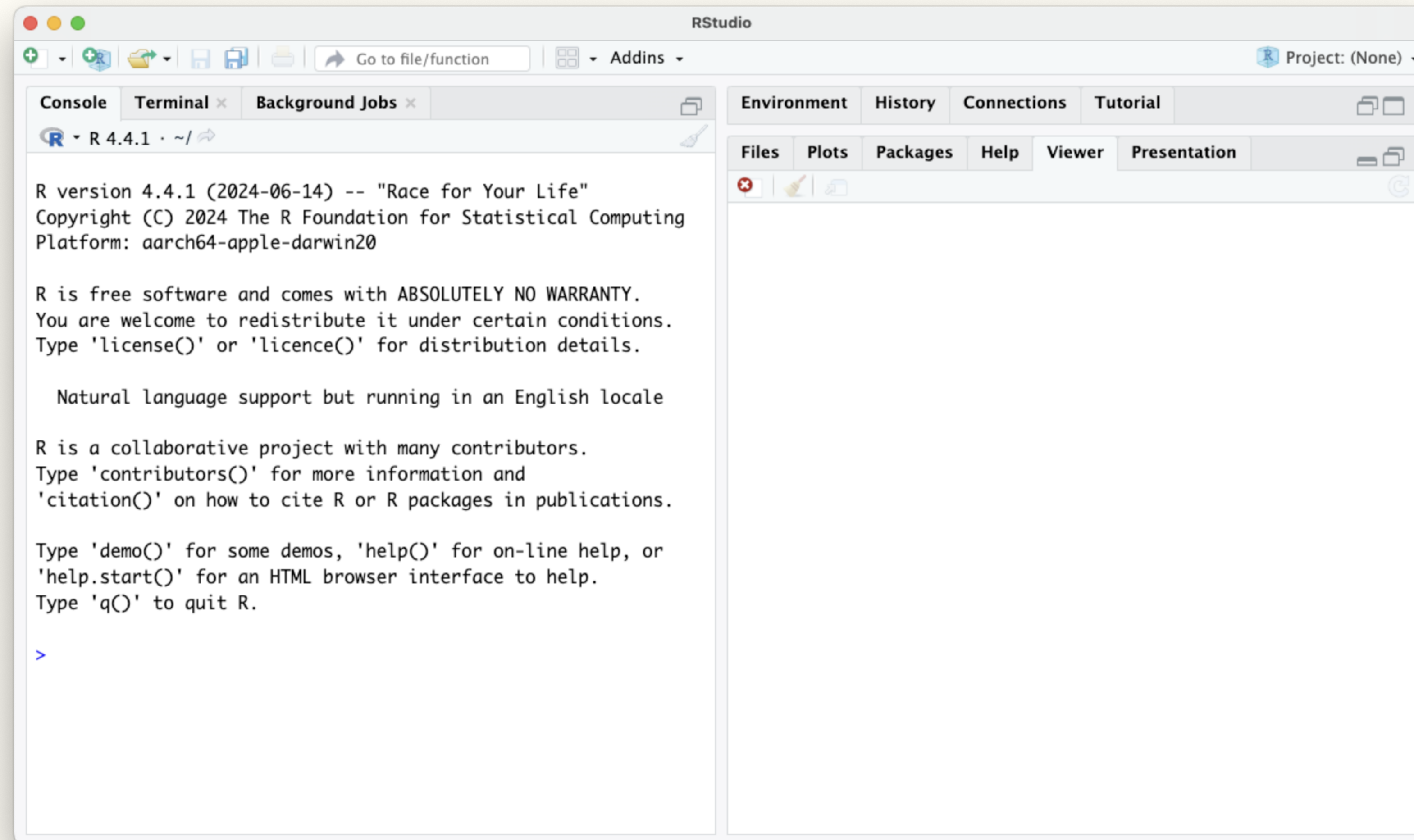
Heather Mattie, PhD

What is RStudio?

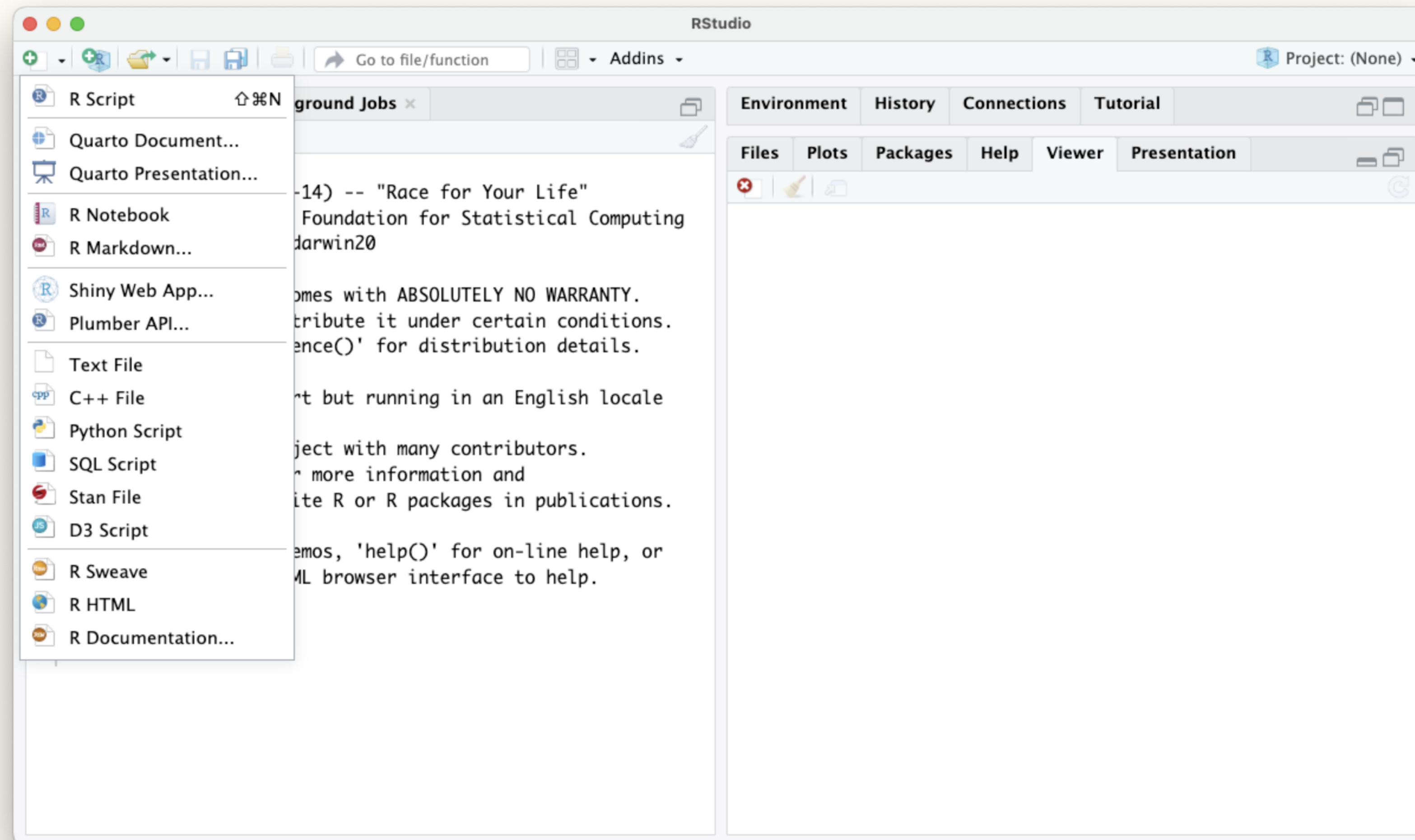
- **RStudio** is an integrated development environment (IDE) specifically created for R
 - RStudio desktop
 - User-friendly interface



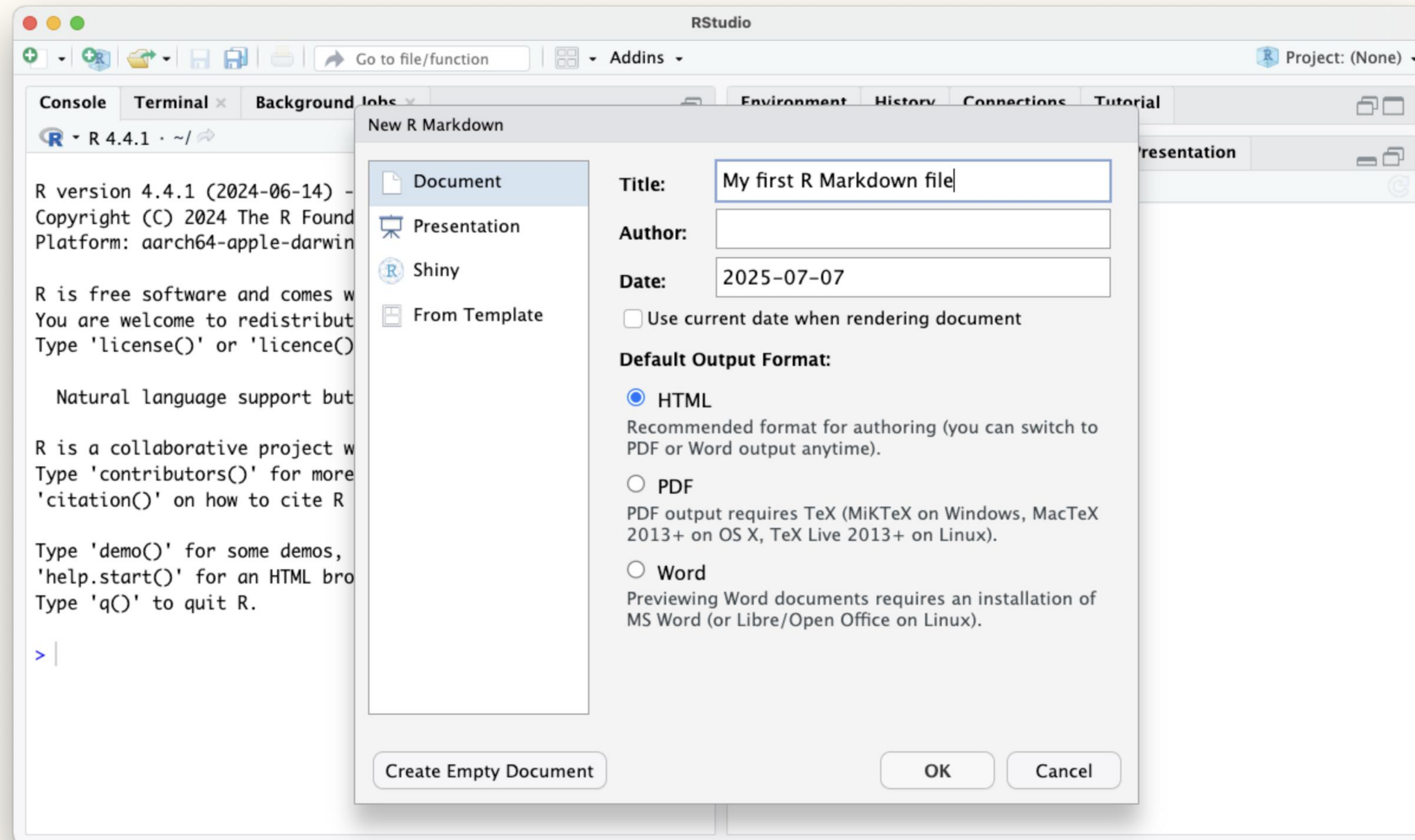
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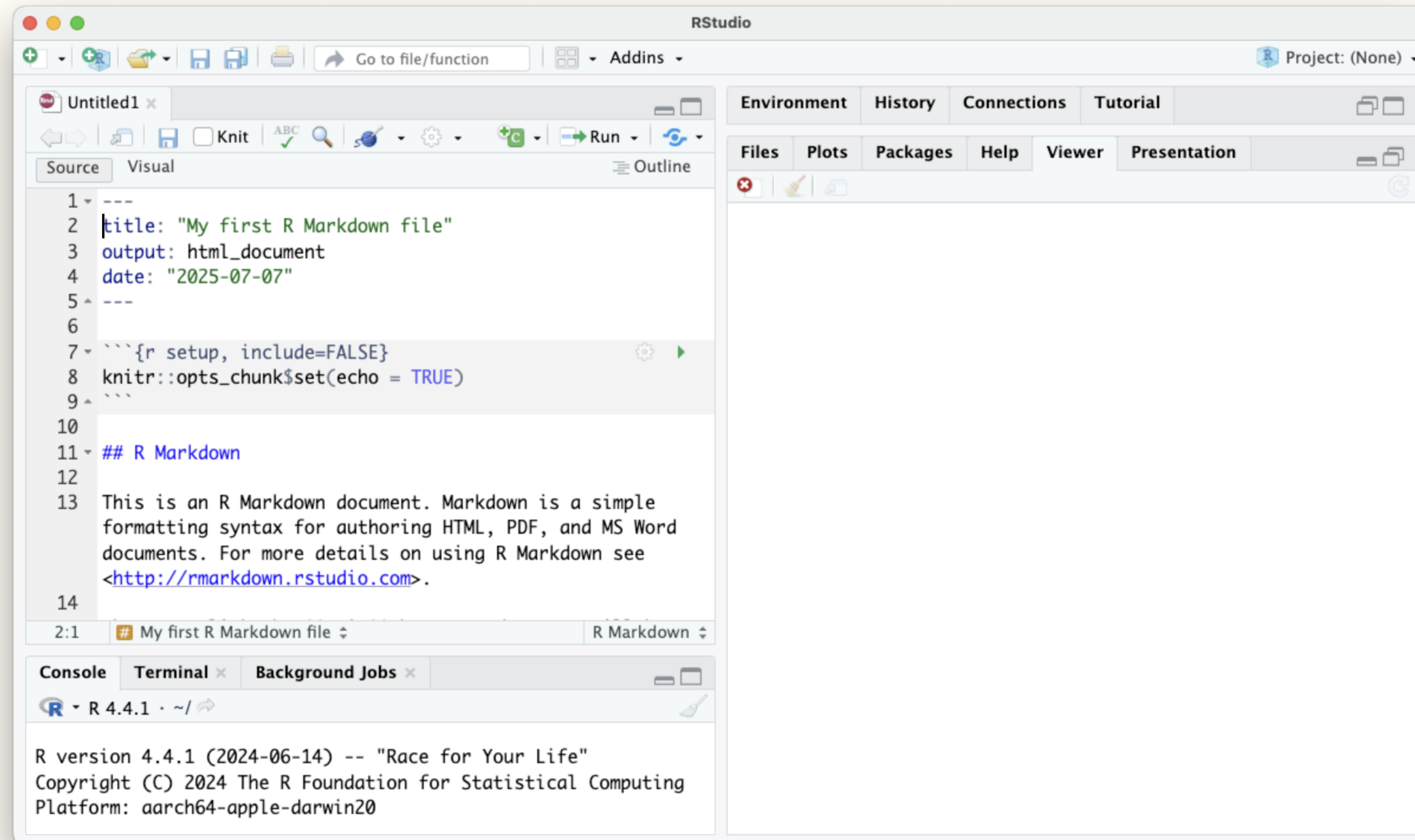
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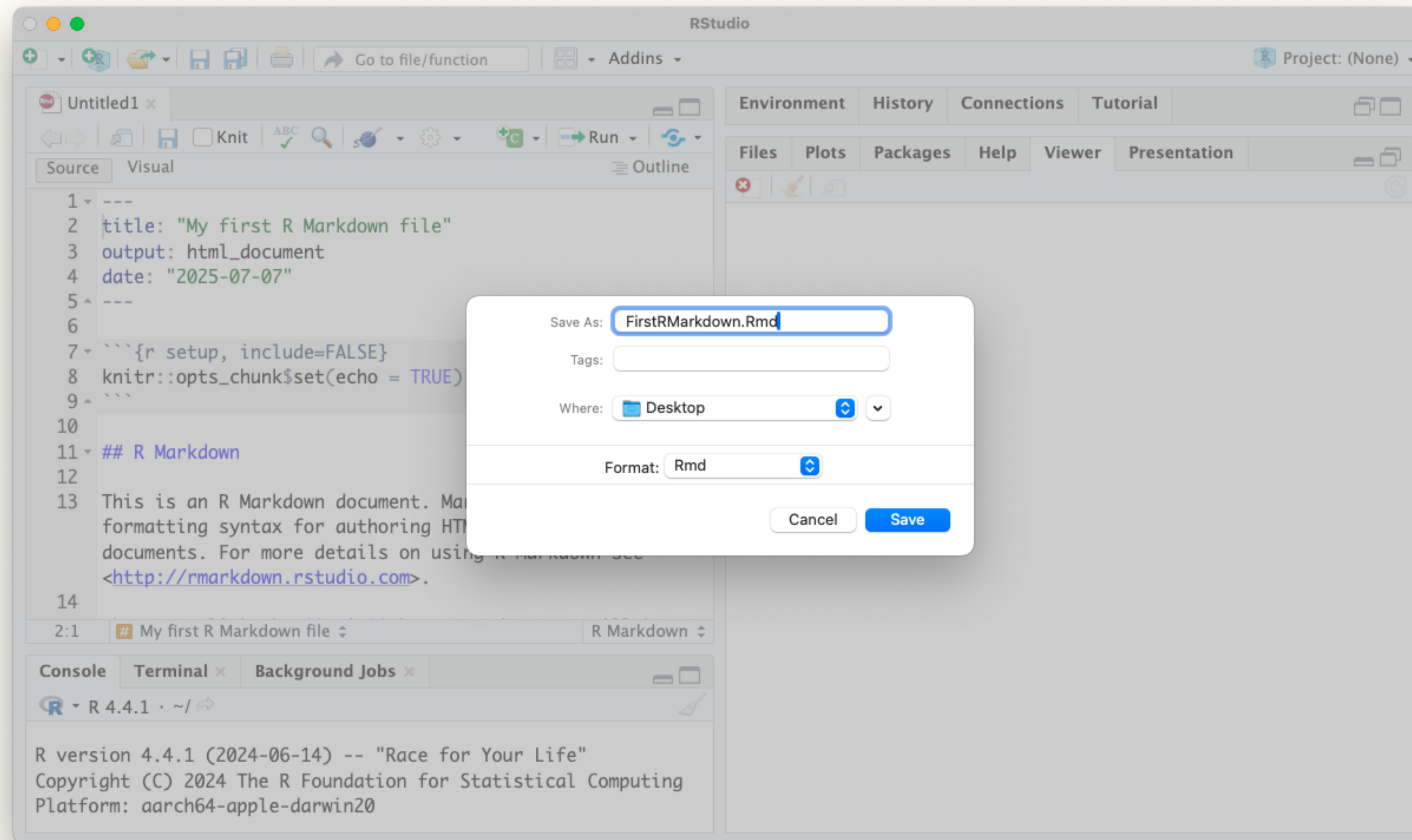
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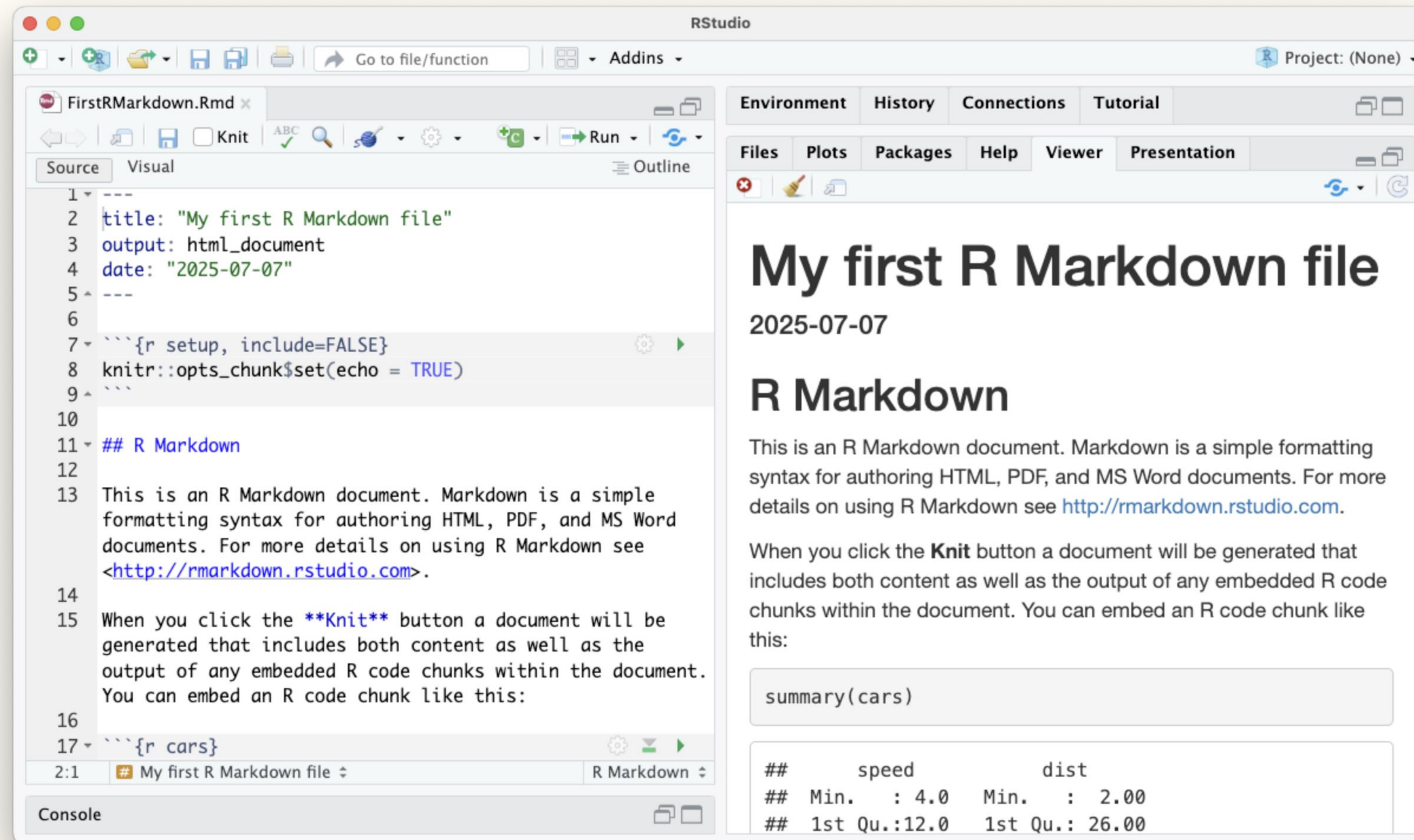
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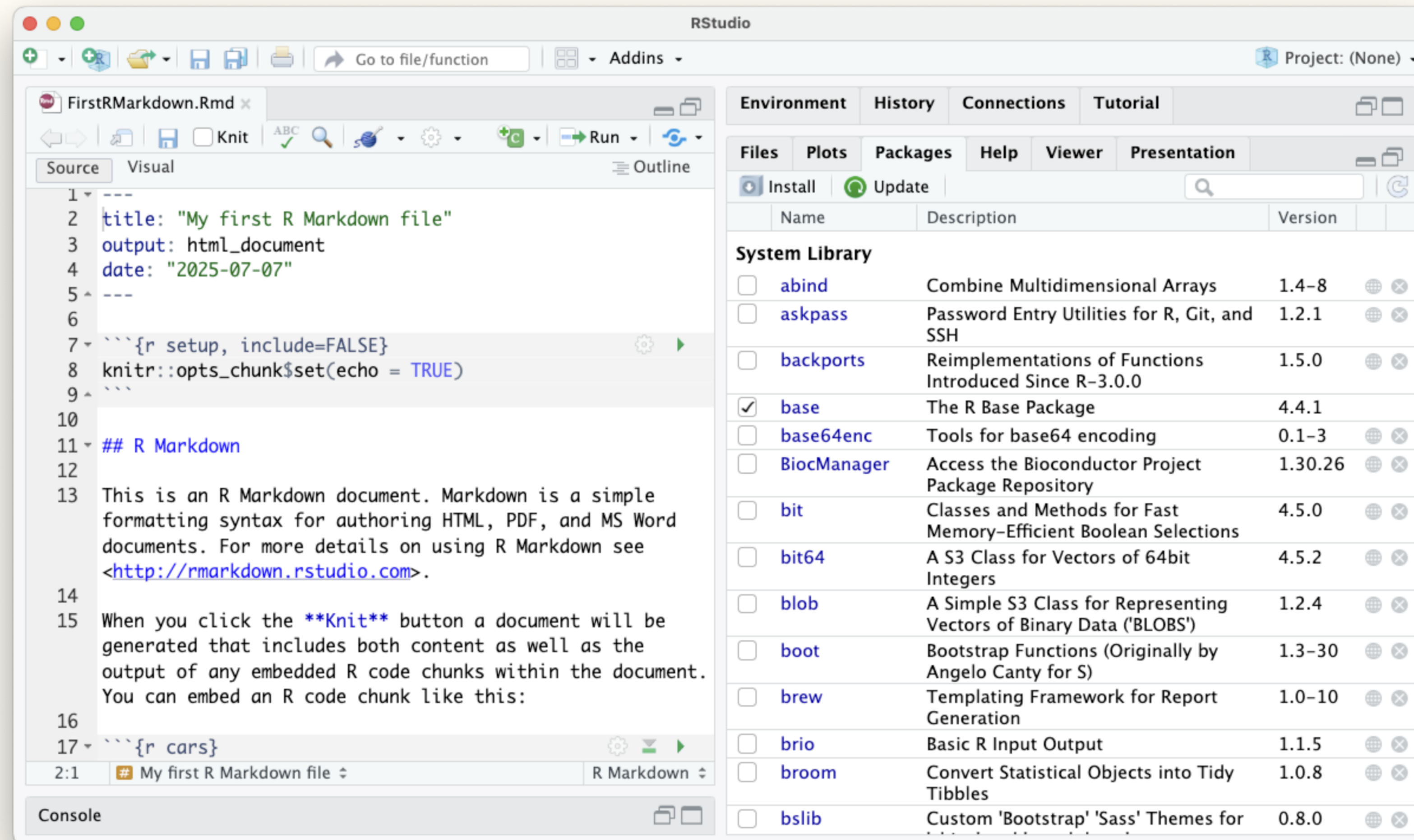
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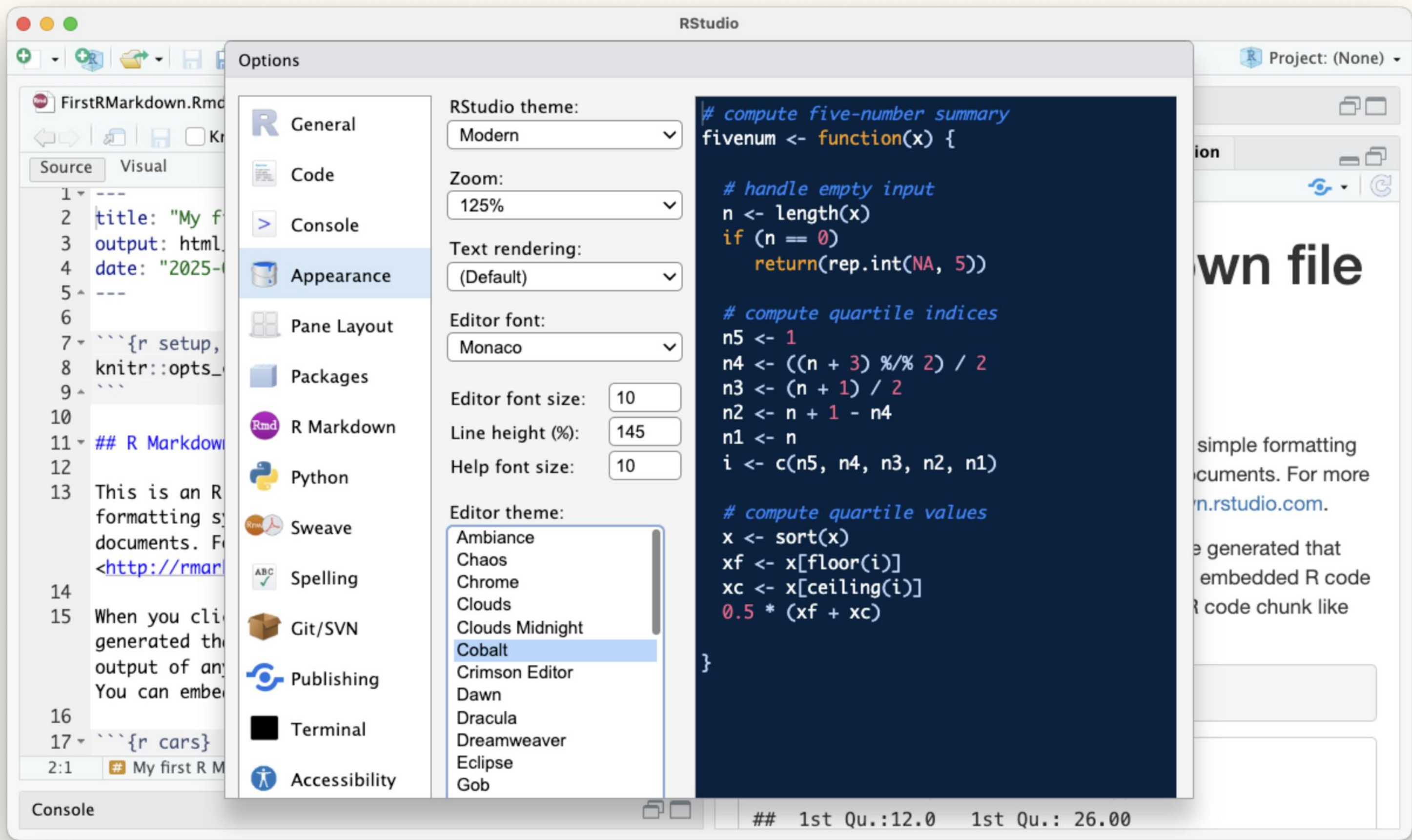
Layout



Layout



Appearance



Summary

- RStudio is an integrated development environment specifically created for R
- RStudio provides a user-friendly interface
- RStudio windowpanes showcase code, an opened file, output, etc.
- The appearance of RStudio can be changed to your preferred style